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Brief report

Gastric dysmotility in patients with major depression

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Abstract

Background: Depressed patients frequently complain about vegetative symptoms. The aim of the present study was to evaluate gastric electrical activity in patients suffering from major depression in relation to their symptoms.
Methods: Electrogastrography (EGG) was performed before and after a test meal ingestion in 21 patients suffering from major depression and control subjects. A structured interview was used to assess the severity of clinical as well as abdominal symptoms.
Results: Patients presented with significantly elevated proportions of tachygastria. Significant differences were found for the parameter arrhythmia, the coefficient of dominant frequency and slow wave after the test meal ingestion. A positive correlation was found between tachygastria and ANS score as well as tachygastria and the item "sweating".
Limitations: Further studies are warranted to include more patients and to investigate interaction with specific antidepressive drugs.
Discussion: This is the first study demonstrating an increased amount of tachygastria in patients suffering from major depression which might be caused by increased sympathetic modulation. The correlation of tachygastria with the amount of gastric symptoms underlines the clinical significance of this finding.
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1. Introduction

Major depressive disorder (MDD) is known to influence the equilibrium of the autonomic nervous system (Yeragani et al., 2002; Moser et al., 1998; Bar et al., 2004). These patients often complain about a variety of vegetative symptoms such as disturbances of appetite, digestion, meteorism, sweating or constipation (Davidson and Turnbull, 1986). Anxiety is a common symptom of depression and is well known to occur in patients with irritable bowel syndrome (IBS) (Lydiard, 2001).

Autonomic dysfunction has been demonstrated in many studies investigating the influence of MDD on cardiac regulation (Bar et al., 2004; Boettger et al., 2008). A decreased parasympathetic and increased sympathetic function has been frequently suggested (Bar et al., 2004; Voss et al., 2006; Boettger et al., 2008). Yeragani et al. have found increased beat-to-beat QT variability in

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patients with depression as well as anxiety and it seems to be related to sympathetic function (Yeragani et al., 2000).

The enteric nervous system has not been studied in more detail in patients suffering from MDD although constipation is found in almost a third of depressed patients (Garvey et al., 1990).

Former reports have found some evidence of a prolongation of gut transit time in these patients (Gorard et al., 1996). Similarly, a high proportion of patients with intractable constipation suffered from a previous or current affective disorder (Dykes et al., 2001). Wu and colleagues have shown that dyspeptic patients with slightly dysthymic symptoms exhibit more dysrhythmic events of the gastric pacemaker by means of noninvasive electrogastrography (EGG; Wu et al., 2003). Several investigators have used the EGG to study conditions that have delayed gastric emptying such as diabetic gastroparesis or dyspepsia and have found that these are associated with tachygastria (Chen et al., 1993; Chen and McCallum, 1994a). Increased bradygastria indicated enhanced parasympathetic modulation in anorexia nervosa (Ogawa et al., 2004).

In the present study, we have investigated 21 unmedicated patients suffering from MDD by means of EGG to look for evidence of autonomic dysfunction within the enteric nervous system. We hypothesized that there is an increase in gastric sympathetic function in patients with MDD.

2. Material and methods

2.1. Subjects

We recruited 21 patients with major depressive disorder (MDD) and compared the EGG parameters of these patients with 21 healthy volunteers (sex- and age-matched). Patients admitted to our inpatient unit were diagnosed by a staff psychiatrist. To include patients into the study the patients had to meet DSM-IV criteria for MDD. All patients had not taken antidepressants for at least 8 weeks prior to hospital admission. Diagnosis was confirmed by means of a structured clinical interview for DSM-IV Axis I disorders (SCID, 1997). Any additional or interfering psychiatric condition (e.g. anxiety disorder) led to the exclusion of patients. The Hamilton Depression Rating Scale (HAM-D-21; Hamilton, 1960) and the Beck Depression Inventory (BDI; Beck et al., 1961) were used to quantify the severity of depression.

The nature of the study required to exclude any subject with a history or symptom of peptic ulcer disease, gastroesophageal reflux disease, irritable bowel syndrome, symptoms of gastritis, diabetes mellitus or any further interfering medical or neurological condition as well as a previous gastrointestinal surgery. None of the participants was in receipt of any medication. In addition, control subjects had to have unremarkable scores on the HAM-D-21 (<7 points), BDI (<5 points) and autonomic symptom score (ANS score <3 points).

We obtained an ANS score, which includes the following symptoms: feeling of fullness, heartburn, nausea, vomiting, dry mouth, diminished appetite, constipation, diarrhea, palpitation, sweating, back pain, muscle pain, headache and abdominal pain. In addition, we assessed (0 = no; 1 = mild; 2 = medium; 3 = severe) the severity of these symptoms (Table 1).

All participants gave written informed consent to a protocol approved by the local Ethics Committee.

2.2. Electrogastrography

EGG (Medtronic, Minneapolis, USA) measures the activity of the gastric pacemaker in the corpus of the stomach using cutaneous electrodes (Chen et al., 1995). The signal contains slow waves (electrical control activity) as well as spike activity.

Gastric slow waves represent the dominant frequency obtained from spectral analyses corresponding with frequency of phasic gastric contractions (Chen and McCallum, 1994a).

While the percentage of slow waves reflect the regularity of gastric myoelectrical activity (normogastria in %, 2.4–3.6 cycles/min), the superimposed spikes are associated with the magnitude of contractions (Smout et al., 1980; Chen and McCallum, 1994a). Gastric dysrhythmia includes bradygastria (0.5–2.4 cycle/min), tachygastria (3.7–9.0 cycle/min) and arrhythmia (Chen and McCallum, 1994a). Bradygastria, is mainly associated with vagal activity (Qian et al., 2003). Tachygastria arises due to additional and external pacemaker activity located in the antrum of the stomach propagating retrograde to the corpus (Qian et al., 2003). An increased proportion of tachygastria and arrhythmia might be associated with increased sympathetic modulation (Chen et al., 1993).

The instability coefficient of dominant frequency (ICDF) and the dominant power (ICDP) are indicators of the stability of gastric electrical peak and reflect changes in gastric slow waves (Diamanti et al., 2003, Chen and McCallum, 1994b).

The percentage of slow-wave coupling is a measure of the coupling between two EGG channels and it is defined as the ratio between the number of time segments during which the recorded slow waves were coupled and the total number of segments (Chen et al., 1999; Xu et al., 2004).

2.3. Procedure of EGG recordings

Patients were studied in the morning after an overnight fast. The EGG was recorded for 30 min in fasting state and thereafter for another 30 min during standard meal digestion (200 ml Clinutren®, 300 kcal, Vevey, Switzerland). All electrodes were positioned to follow the antral axis of the stomach. Channel one was placed over the fundus region and channel four over the antrum region. The best results were obtained from channel 3, which is positioned above the gastral pacemaker.

2.4. Statistical analysis

The natural logarithm of all values was calculated to generate normal distribution. A repeated measure MANOVA was used with controls versus patients as the grouping factor and before and after food parameters as the repeated measure. Significant effects were followed up by one way ANOVA for the single parameters normogastria, tachygastria, bradygastria, arrhythmia, coefficient of dominant frequency (ICDF), dominant power (ICDP) and slow waves.

Post-hoc *t*-tests were used to examine differences between patients and controls (Fig. 1). In addition, we correlated symptoms of depression and the ANS score with the EGG parameters using the Spearman rank correlation.

3. Results

The MANOVA revealed a significant GROUP effect (F (34,7) = 2.5; p = 0.046) indicating an overall difference between patients and controls (Fig. 1 and Table 1). Tachygastria was the only parameter that was significantly different in patients before and after the

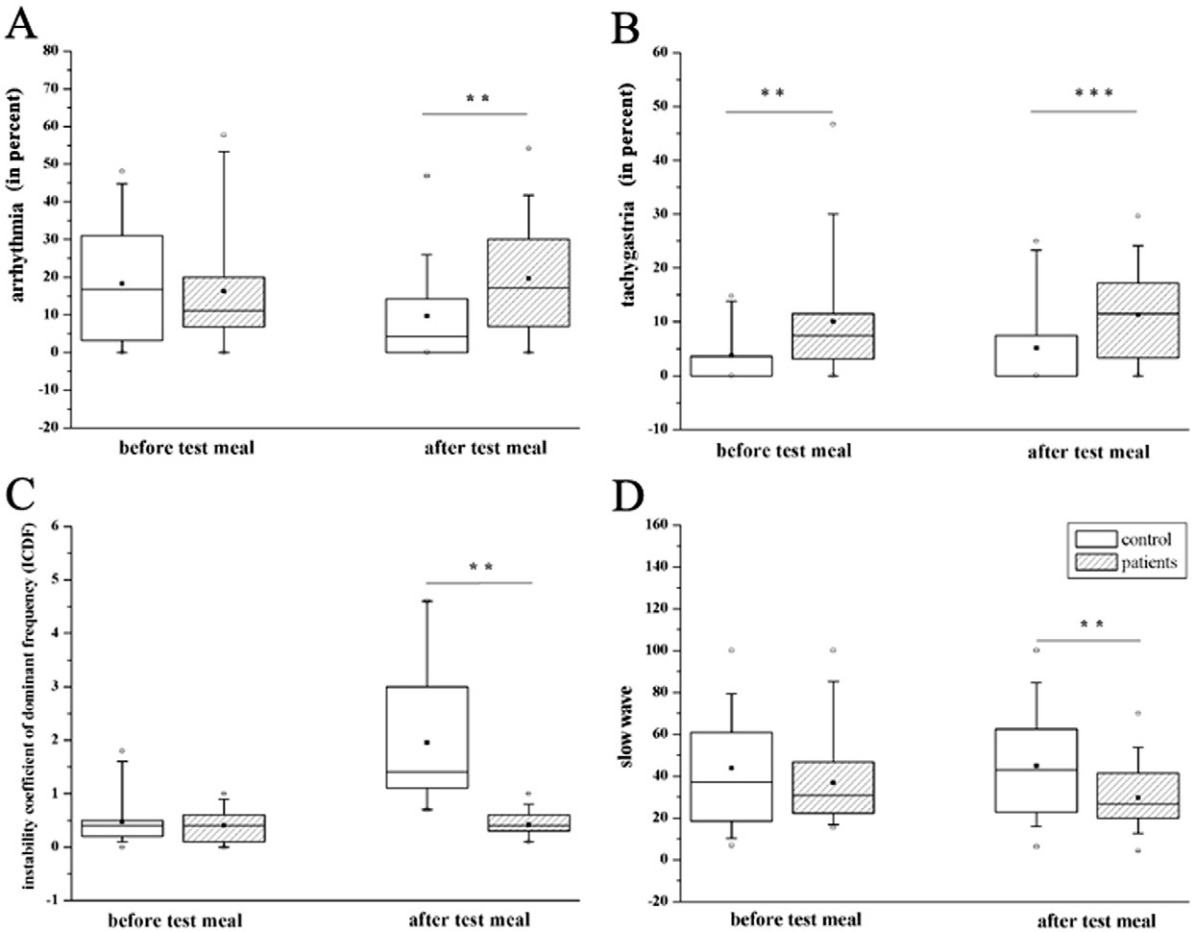

Fig. 1. The significant increase of arrhythmia after the test meal ingestion in depressed patients is presented in A. Tachygastria (B) is elevated before and after ingestion of the test meal indicating increased sympathetic modulation. ICDF (C) and slow-wave (D) parameters were significantly decreased after test meal ingestion indicating disturbed electrical propagation.

Table 1
Demographic data of participants

Parameter	Controls	Patients
	$n=21$	$n=21$
Age (years)	38.86±15.95	38.27±16.16
Sex (F = female; M = male)	F=16; M=5	F=16; M=5
Smokers	$n=7$	$n=9$
Body mass index	22.80±3.16	22.92±4.58
HAM-D	3.4±1.4	24.86±8.034
BDI	3.9±1.7	22.30±10.00
Autonomic symptom score	<2	7.10±4.75
Normogastria before test meal (in %)	69.3±11.4	67.4±5.04
Normogastria after test meal (in %)	74.3±4.3	56.39±5.01
Bradygastria before test meal (in %)	13.3±3.2	7.9±1.3
Bradygastria after test meal (in %)	10.3±2.5	11.3±2.6
ICDP before test meal	0.95±0.06	1.51±0.23
ICDP after test meal	1.85±0.28	1.36±0.15

Data are presented as mean±standard deviation. HAM-D = Hamilton Depression Rating Scale; BDI = Beck's Depression Inventory; ICDP instability coefficient of dominant power.

test meal ingestion (F = 13.7; $p<0.001$; Fig. 1B). Although we have observed only a trend for TIME ($p<0.052$) for both groups, a significant GROUP × TIME interaction (F (34,7) = 3.74; p = 0.002) revealed differential changes of gastric parameters of patients and controls after ingestion of the test meal. In particular, one way ANOVA revealed a GROUP × TIME interaction for arrhythmia (F = 10.0; $p<0.003$; Fig. 1A), normogastria (F = 4.67; p = 0.037), the coefficient of dominant power (F = 8.75; $p<0.005$) as well as a trend for slow waves (F = 3.32; $p<0.07$; Fig. 1D) and the coefficient of dominant frequency (F = 3.50; $p<0.069$; Fig. 1C).

In addition, a positive correlation was found between the ANS score and the percentage of tachygastria (r = 0.764; $p<0.001$) prior to ingestion of the test meal. Similarly, we observed a significant correlation between the item "sweating" of the ANS score and tachygastria (r = 0.685; $p<0.003$).

4. Discussion

The main finding of our study is the significantly increased amount of tachygastria. This suggests elevated sympathetic modulation in these patients. Tachygastria is assumed to arise due to additional external pacemaker activity (Jin et al., 2007; Hu et al., 1991) and correlates with gastric hypomotility (Qian et al., 2003; Chen et al., 2005). In previous studies, the autonomic function in depressed patients was evaluated by measuring the heart rate variability (HRV) (Agelink et al., 2001; Bar et al., 2004; Boettger et al., 2008). It is therefore interesting that we report increased sympathetic modulation in the gastrointestinal system. We suppose that tachygastria is caused by central autonomic dysfunction influencing sympathetic modulation in the gastrointestinal tract in these patients. Studies on the central modulation of sympathetic tone in depression are very much needed.

We also found a positive correlation of gastric symptoms with tachygastria in our patients which points to the relation of sympathetic function and abdominal symptoms. The central modulation of sympathetic function in depression is also indicated by the positive correlation of tachygastria and the subjective rating of sweating, since the sweating depends on sympathetic function as well (Vetrugno et al., 2003).

Elevated proportions of tachygastria as well as arrhythmia are mainly found in patients with dyspepsia (Chen and McCallum, 1994a; Pfaffenbach et al., 1997, 1998). The significant increase of arrhythmia after test meal ingestion is the second important finding, which is thought to be related to the absence of gastric contractions (Chen et al., 2005). Changes in EGG rhythm from normogastria to tachygastria and arrhythmia are thought to cause epigastric distress and nausea (Koch and Stern, 1996; Walsh et al., 1996; Ogawa et al., 2004).

The measurement of slow-wave activity allows the assessment of the propagation of abdominal contractions and their frequency (Chen et al., 1999). The reduced slow-wave activity is a further indication of gastric dysmotility. In addition, we have observed that the instability coefficient of the dominant frequency (ICDF) is not reduced after test meal ingestion. This might again be associated with feelings of fullness and abdominal distention (Ogawa et al., 2004; Tack et al., 1998). However, these results need to be viewed with

some caution since we excluded abdominal pathology by clinical methods only. Although it is unlikely that our depressed patients suffered from an underlying medical disorder, we can not exclude this completely in this study.

In conclusion, our findings suggest that major depression is associated with gastric dysrhythmia possibly caused by increased sympathetic modulation. Further studies are warranted to gain more insight into the complex interaction of autonomic enteric function and major depression.

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Conflict of interests

None to declare.

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